Ingredio

Find recipes from what you already have

Mobile Applications · THWS / TAMK · 2026

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What is Ingredio?

The problem

You open the fridge, see some ingredients, and have no idea what to cook. You either waste food or order takeout.

Our solution

Select the ingredients you have. Ingredio instantly fetches and ranks real recipes by how many of your ingredients they need — no guesswork, no waste.

7	Screens
9	Riverpod providers
10+	Tests
10	Packages used

Key Features

Smart Pantry

Search, select & quantity-tag up to 30 ingredients with debounced live search

Recipe Discovery

Recipes ranked by match count — the more ingredients you have, the higher it ranks

Favorites

Save any recipe locally via Hive for offline access

Profile & Onboarding

Name registration, editable profile, recipe collections, dietary prefs

Offline Support

Full Hive caching — ingredient list and recipes work without internet

Share Recipes

Native share sheet to send recipe details to any app

Architecture

Clean Architecture — 3 decoupled layers

Presentation

screens/ · widgets/ · providers/

Flutter widgets + Riverpod AsyncValue

Domain

entities/ · usecases/ · interfaces/

Pure Dart — zero Flutter dependencies

Data

api/ · local/ · repositories/

MealDB API (Dio) + Hive local cache

Dependency Injection via get_it — domain interfaces resolved to data implementations at startup

State Management

flutter_riverpod 2 — AsyncValue for every async operation

Provider	Type	Purpose
`recipesProvider`	`FutureProvider`	Matched recipes list
`recipeDetailProvider`	`FutureProvider.family`	Single recipe by ID
`ingredientsListProvider`	`FutureProvider`	Full ingredient list from API/cache
`ingredientsProvider`	`StateNotifierProvider`	Selected ingredients set
`ingredientQuantitiesProvider`	`StateNotifierProvider`	Quantity per ingredient
`favoritesProvider`	`StateNotifierProvider`	Persisted favourites
`connectivityProvider`	`FutureProvider`	Network state
`userProfileProvider`	`StateNotifierProvider`	Registered user name
`hiveDatabaseProvider`	`Provider`	Hive singleton

App Screens

Splash

Reads Hive on startup; routes to Register or Main

Register

Name input with validation; persisted to Hive; dark/light theme

Pantry

Ingredient list, live search, selection, quantities

Discover

Recipes ranked by match count + client-side search filter

Recipe Detail

Instructions, measures, YouTube link, share button

Favorites

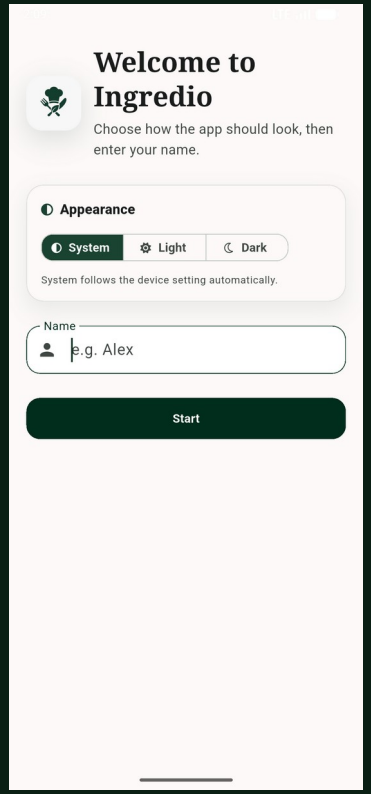
Offline recipe list from Hive

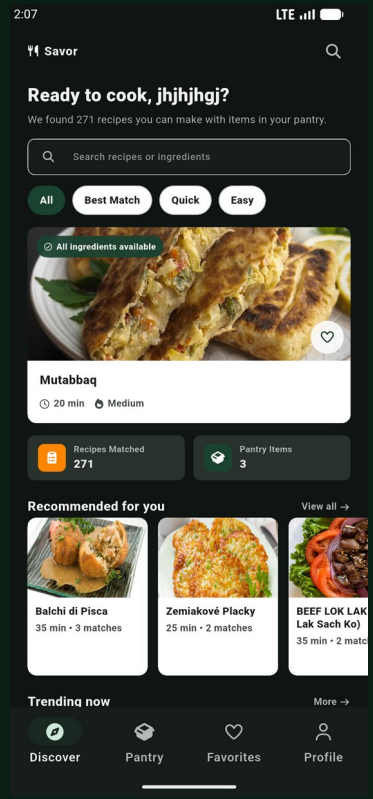
Profile/Settings

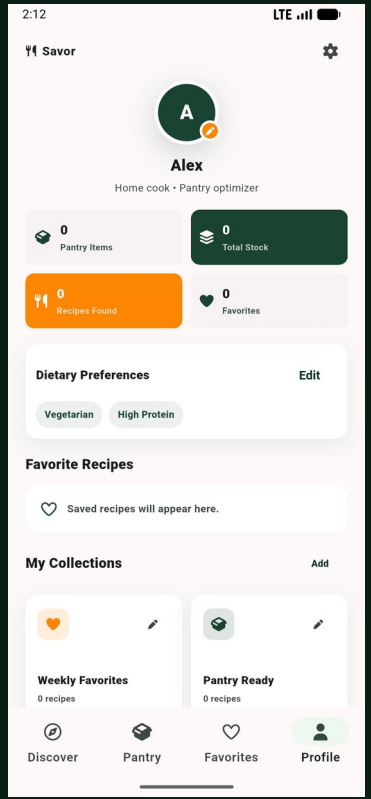
User name, collections, prefs, logout / delete all dark/light theme

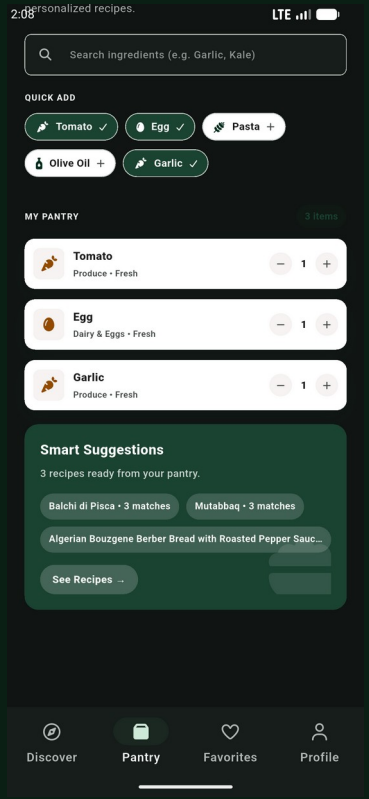
Navigation: Navigator.push(MaterialPageRoute) · Bottom tab: IndexedStack with lazy loading · iOS: Custom CupertinoButton bar

App Screens


Welcome to Ingredio
Choose how the app should look, then enter your name.
Appearance
System
Light
Dark
System follows the device setting automatically.
Name
e.g. Alex
Start


Savor
Ready to cook, jhjhjhgj?
We found 271 recipes you can make with items in your pantry.
Search recipes or ingredients
All
Best Match
Quick
Easy
All ingredients available
Mutabbaq
20 min
Medium
Recipes Matched
271
Pantry Items
3
Recommended for you
View all →
Balchi di Pisca
35 min • 3 matches
Zemiakové Placky
25 min • 2 matches
Trending now
More →
Discover
Pantry
Favorites
Profile


Savor
A
Alex
Home cook • Pantry optimizer
0
Pantry Items
0
Total Stock
0
Recipes Found
0
Favorites
Dietary Preferences
Edit
Vegetarian
High Protein
Favorite Recipes
Saved recipes will appear here.
My Collections
Add
Weekly Favorites
0 recipes
Pantry Ready
0 recipes
Discover
Pantry
Favorites
Profile


personalized recipes.
Search ingredients (e.g. Garlic, Kale)
QUICK ADD
Tomato
Egg
Pasta
Olive Oil
Garlic
MY PANTRY
Tomato
Produce • Fresh
Egg
Dairy & Eggs • Fresh
Garlic
Produce • Fresh
Smart Suggestions
3 recipes ready from your pantry.
Balchi di Pisca • 3 matches
Mutabbaq • 3 matches
Algerian Bouzgene Berber Bread with Roasted Pepper Sauc...
See Recipes →
Discover
Pantry
Favorites
Profile

Data Flow

AsyncValue pattern — every screen handles all 3 states:

`loading:` CircularProgressIndicator shown centrally

`error:` Error message + Retry button, calls ref.invalidate()

`data:` Content rendered; empty state handled gracefully

Testing

flutter test — all passing

Unit Tests

ingredient_icons_test.dart

IngredientIcons.forName() — icon & category mapping for poultry, spices, unknowns

get_recipes_by_ingredients_test.dart

Empty input returns early · Deduplication of ingredients · onlyBasicInfo flag passthrough

get_recipe_details_test.dart

GetRecipeDetails delegates to repository correctly

Widget Tests

recipes_list_screen_test.dart

Renders featured + recommended sections; search bar filters results

ingredients_screen_test.dart

Renders ingredient list; selection toggle

recipe_detail_screen_test.dart

Renders recipe name, instructions, action buttons

profile_screen_test.dart

Renders user name and profile sections

Tech Stack & Packages

Package	Why we use it
`flutter_riverpod ^2.0.0`	State management — providers, notifiers, AsyncValue
`get_it ^7.2.0`	Service locator / dependency injection
`dio ^5.0.0`	HTTP client — MealDB API calls
`hive + hive_flutter`	Lightweight local NoSQL — offline caching
`cached_network_image`	Efficient URL image loading + disk cache
`connectivity_plus`	Detect network state
`share_plus`	Native share sheet
`url_launcher`	Open YouTube links in system browser
`font_awesome_flutter`	Icon set used throughout the UI

Thank you!

Questions welcome

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github.com/soffffises/Ingredio

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